

HDMI In Consumer Electronics



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High-definition multimedia interface, or HDMI, is one of those technologies that have revolutionised consumer electronics audio and video display and media players. HDMI technologies are being used since more than a decade now. Predecessor of HDMI was DVI interface, which can still be seen at times in today's modern world. DVI was mainly meant for video displays.

HDMI protocol runs at high-speed differential signaling with maximum clock rate of 600MHz. The transition minimised differential signaling clock rate varies

from 25MHz to 600MHz. If any resolution has a lower clock rate than 25MHz, the source device applies the pixel clock repetition mechanism to meet 25MHz or more clock frequency.

Compared to DVI, HDMI can support a wide range of video resolutions and audio codecs and bit rates. It can support deep colours up to 36 bits, and hence can provide 2^{36} different colours in TV. Therefore we can also say that HDMI can provide true colours in a TV.

HDMI is a digital interface that exists in almost all modern consumer electron-

TABLE I
HDMI SUPPORT WITH DIFFERENT VERSIONS

HDMI version	1.0	1.1	1.2 1.2a	1.3 1.3a 1.3b 1.3b1 1.3c	1.4 1.4a 1.4b	2.0 2.0a
sRGB	Yes	Yes	Yes	Yes	Yes	Yes
YCbCr 4:2:2/4:4:4	Yes	Yes	Yes	Yes	Yes	Yes
Eight-channel LPCM, 192kHz, 24-bit audio capability	Yes	Yes	Yes	Yes	Yes	Yes
Blu-ray disk and HD DVD video and audio at full resolution	Yes	Yes	Yes	Yes	Yes	Yes
Consumer electronic control	Yes	Yes	Yes	Yes	Yes	Yes
DVD-audio	No	Yes	Yes	Yes	Yes	Yes
Super audio CD (DSD)	No	No	Yes	Yes	Yes	Yes
Deep colour	No	No	No	Yes	Yes	Yes
xvYCC	No	No	No	Yes	Yes	Yes
Auto lip-sync	No	No	No	Yes	Yes	Yes
Dolby TrueHD bitstream capable	No	No	No	Yes	Yes	Yes
DTS-HD Master Audio bitstream capable	No	No	No	Yes	Yes	Yes
Updated list of CEC commands	No	No	No	Yes	Yes	Yes
3D over HDMI	No	No	No	No	Yes	Yes
Ethernet channel	No	No	No	No	Yes	Yes
Audio return channel (ARC)	No	No	No	No	Yes	Yes
4K resolution at 30Hz	No	No	No	No	Yes	Yes
4K resolution at 60Hz	No	No	No	No	No	Yes
1920x1080 resolution at 120Hz	No	No	No	No	Yes	Yes
Rec. 2020 colour space	No	No	No	No	No	Yes
YCbCr 4:2:0	No	No	No	No	No	Yes
32-channel audio	No	No	No	No	No	Yes
1536kHz total audio	No	No	No	No	No	Yes
Four audio streams	No	No	No	No	No	Yes
Two video streams (dual view)	No	No	No	No	No	Yes
21:9 aspect ratio	No	No	No	No	No	Yes



Fig. 1: HDMI logo